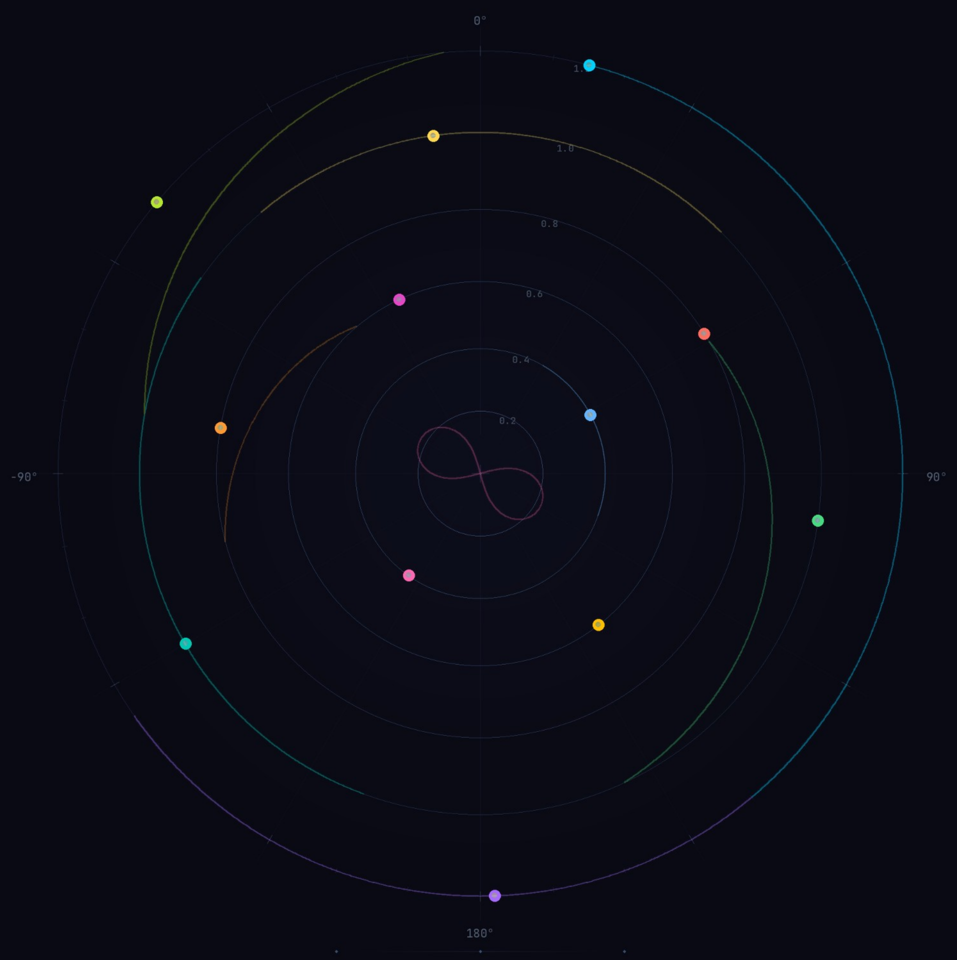


OpenSpatialDelay

v1.0 User Manual

Spatial Media Lab

Contents

1 Quick Start

This page gets you from installation to first sound in under two minutes. For detailed explanations, see the sections that follow.

1.1 Installation

Download the latest release for your platform from the [GitHub Releases page].

macOS: Run the .pkg installer. It places the AU and VST3 plugins in `~/Library/Audio/Plug-Ins/` automatically.

Windows: Run the .exe installer. The VST3 is installed to `C:\Program Files\Common Files\VST3\`.

1.2 Load in Your DAW

Insert OpenSpatialDelay as an effect (insert/send) on any audio or bus track. For binaural (headphone) output, use a stereo track. For surround formats, match the track channel count to your target format.

1.3 Interface Overview

The interface has four regions:

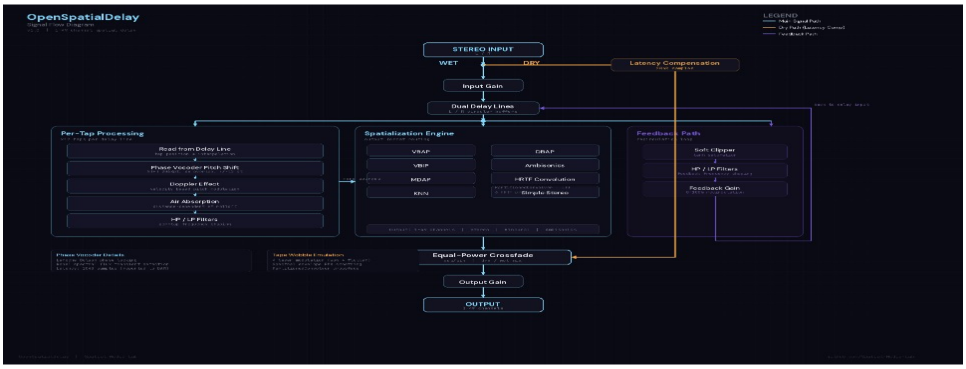
A) Header Bar (top) — presets, output format, and algorithm selection.

B) Spatial Map (left) — top-down view of 3D space where you position delay taps.

C) Bottom Panel (bottom-left) — per-tap controls for the selected object.

D) Right Panel (right) — global controls for delay, tone, modulation, and mix.

1.4 Signal Flow



Audio flows through five stages: **Input** is mixed with feedback and written to the delay buffer. Each enabled **tap** reads from the buffer at its sequential position and applies pitch shifting. Each tap is then **spatialized** to its 3D position. The **feedback** loop reads mono from the end of the tap chain, filtering and soft-clipping before feeding back. Finally, dry and wet signals are mixed at the **output**. See Controls Reference for parameter details.

1.5 Format Quick Pick

Choose your output format based on your workflow:

Goal	Output Format	Algorithm / Profile
Headphone mixing	Binaural	Select HRTF profile (try Studio Reference)
Stereo monitoring	Stereo	Choose mic mode (Equal Power, XY, MS, etc.)
Dolby Atmos (7.1.4)	7.1.4 Atmos	VBAP or MDAP
Ambisonics pipeline	FOA through 6OA	Ambisonics Encode (automatic)
Standard surround	5.1 / 7.1	VBAP (focused) or DBAP (diffuse)

2 What is OpenSpatialDelay?

OpenSpatialDelay is a **spatial delay effect** where each delay tap is positioned in 3D space. Think of a classic stereo ping-pong delay — but instead of bouncing between left and right, the echoes travel through up to 12 spatial positions around the listener.

Each tap reads from a shared delay line at sequential intervals: Tap 1 plays at 1× the delay time, Tap 2 at 2×, and so on. Every tap has its own position (azimuth, elevation, distance), trajectory animation, pitch shift, and Doppler amount.

The output is rendered through one of **five rendering paths**: direct binaural HRTF convolution for headphones, stereo mic simulation, discrete surround speaker panning, or Ambisonics spherical harmonics encoding. See Output Formats & Algorithms for full details.

Key Terminology

Azimuth: Horizontal angle around the listener. 0° = directly in front, +90° = left, -90° = right, ±180° = behind.

Elevation: Vertical angle. 0° = ear level (horizon), +90° = directly above, -90° = directly below.

Distance: How far the sound source is from the listener (0.1 = very close, 3.0 = far away). Affects volume attenuation and air absorption.

HRTF: Head-Related Transfer Function — measurements of how sound changes as it travels around your head and ears. Used to create convincing 3D audio over headphones.

Binaural: Audio rendered for headphone listening, using HRTF processing to simulate 3D positioning.

Surround: Audio rendered to discrete speaker channels (5.1, 7.1.4 Atmos, etc.).

Ambisonics: A format for encoding 3D audio as spherical harmonics. Flexible — can be decoded to any speaker layout later.

2.1 What Makes It Unique

12 independent spatial taps — each positioned anywhere in 3D space, with its own trajectory animation, pitch shift, Doppler amount, and input channel selection.

23 output formats — from binaural headphones to 9.1.6 Dolby Atmos, SpatialMediaLab 13.1, and 6th-order Ambisonics.

8 spatialization algorithms — Constant Power, VBAP, VBIP, MDAP, KNN, DBAP, Ambisonics, and direct binaural HRTF convolution with 6 measured profiles.

Open source — part of the Spatial Media Library suite. Free to use and modify.

3 The Spatial Map

The spatial map is the large visualization on the left side of the plugin window. It shows a top-down view of 3D space with the listener at the center. Each colored dot represents a delay tap positioned in space.

3.1 Coordinate System

The map uses a **polar coordinate system**. The listener sits at the center. Front is at the **top** of the map (marked F). Concentric rings represent distance from the listener.

Direction	Azimuth	Position on Map
Front	0°	Top
Left	+90°	Left
Rear	±180°	Bottom
Right	-90°	Right

Note: The azimuth knob uses **reverse rotation** — turning the knob clockwise moves the tap clockwise on the map. This follows the IEM StereoEncoder convention used in professional spatial audio tools.

3.2 Interaction

Click on a tap dot to select it. The selected tap is highlighted with a glowing ring, and the bottom panel updates to show that tap's controls.

Click and drag a tap to move it in azimuth (rotation) and distance (radial position) simultaneously.

Elevation is controlled via the **ELEV knob** in the bottom panel, not by map interaction.

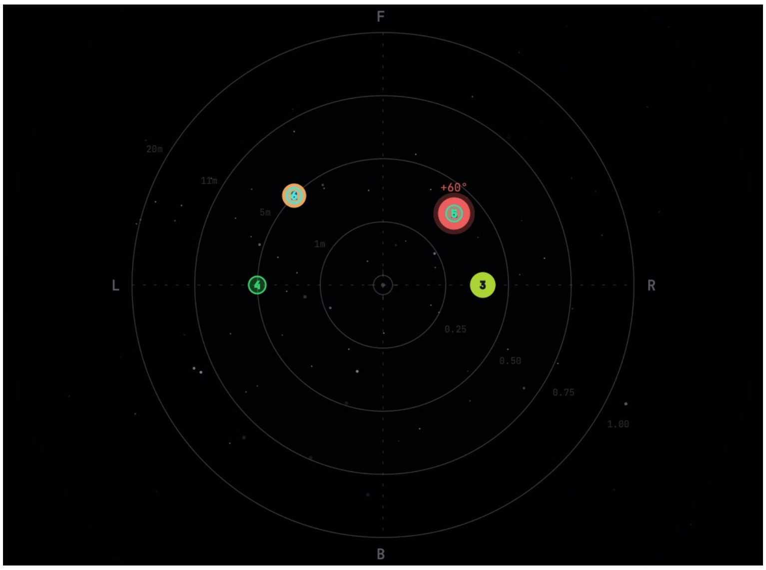
3.3 Elevation Visualization

Although the map is 2D (top-down), elevation is communicated visually through the dot appearance:

Upper hemisphere (elevation > 0°): Tap dots are solid and opaque, slightly larger (+5px). These represent sounds above ear level.

Lower hemisphere (elevation < 0°): Tap dots become transparent (30% opacity) and slightly smaller (-3px). Minimum dot size is 11px at -90°.

At 0° elevation (ear level), dots are at their default size and full opacity.



4 Output Formats & Algorithms

OpenSpatialDelay supports **23 output formats** organized into four categories. The plugin automatically selects the correct rendering path based on the format you choose in the header dropdown.

4.1 Five Rendering Paths

The plugin dispatches audio through one of five peer rendering paths based on the selected output format:

Rendering Path	When Used	Output
Direct Binaural HRTF	Binaural format + SOFA profile	2 ch (L/R)
Simple Binaural (Woodworth)	Binaural + Simple profile selected	2 ch (L/R)
Stereo Variants	Stereo format	2 ch (L/R)
Ambisonics Output	Any Ambisonics format (FOA–6OA)	4–49 ch
Discrete Surround	Any surround format (Quad–9.1.6)	4–16 ch

4.2 Binaural & Stereo Formats

4.2.1 Binaural (2 channels)

The default format. Each tap's 3D position is rendered through **per-source HRTF convolution** — a separate convolver for each active tap, using measured head-related transfer functions from SOFA files (Spatially Oriented Format for Acoustics). This produces highly realistic 3D audio over headphones. Select a profile from the **HRTF Profile** dropdown (see HRTF Profiles).

4.2.2 Stereo (2 channels, 5 modes)

When Stereo is selected, the Algorithm dropdown switches to offer five microphone simulation modes:

Mode	Description
Equal Power Pan	Standard cosine/sine pan law from azimuth angle
VBAP (2-speaker)	Virtual speakers at ±30°, VBAP gain computation
XY Pair	Coincident cardioid microphones at ±45°
MS (Mid-Side)	Mid = cos(azimuth), Side = sin(azimuth)
Blumlein	Crossed figure-8 microphones at ±45°

4.3 Surround Formats

15 discrete surround formats, from Quadraphonic to 9.1.6 Atmos and SpatialMediaLab 13.1. Formats with an LFE channel (marked with .1) derive it as a mono sum of all tap outputs through a 120 Hz low-pass filter at -10 dB. Choose a spatialization algorithm from the Algorithm dropdown.

Format	Channels	LFE	Height Speakers
Quadraphonic	4	No	None
5.0 Surround	5	No	None
5.1 Surround	6	Yes	None
7.0 Surround	7	No	None
7.1 Surround	8	Yes	None
9.1 Surround	10	Yes	None (9 ear-level, ITU-R BS.2051 System H)
Octaphonic	8	No	None (ring at 45° intervals)
5.1.2	8	Yes	2 top
7.1.2 Atmos	10	Yes	2 top
5.1.4 Atmos	10	Yes	4 top
7.1.4 Atmos	12	Yes	4 top
7.1.6 Atmos	14	Yes	6 top
9.1.4 Atmos	14	Yes	4 top + wide speakers
9.1.6 Atmos	16	Yes	6 top + wide speakers
SpatialMediaLab 13.1	14	Yes	8 ear-level + 4 height + 1 zenith

4.4 Ambisonics Formats

Six Ambisonics orders using ACN/SN3D (AmbiX) channel ordering. Decoded to speakers by your DAW or external renderer.

Format	Order	Channels
FOA (1st Order)	1	4
SOA (2nd Order)	2	9
HOA (3rd Order)	3	16
4OA (4th Order)	4	25
5OA (5th Order)	5	36
6OA (6th Order)	6	49

4.5 Spatialization Algorithms

When using surround output formats, the **Algorithm** dropdown lets you choose how tap positions are translated to speaker gains:

Algorithm	Best For	Character
Constant Power	Most surround work (default)	Cosine-distance weighting, wide natural rolloff
VBAP	Precise point sources, production mixing	Focused, sharp localization
VBIP	Off-axis listener stability	VBAP with intensity weighting, smoother
MDAP	Wide, stable spatial images	VBAP + 8 spread sub-sources around main position
KNN	Irregular or curved speaker arrays	3 nearest speakers, distance-weighted, natural
DBAP	Non-standard speaker layouts	Distance-based, warm and diffuse
Ambisonics	Ambisonics workflows, high-order setups	SH encode/decode, smooth panning

Which algorithm should I use?

Start with Constant Power (the default) — it provides smooth, natural panning suitable for most surround work.

For sharper, more focused point sources, switch to VBAP. For wider, more stable sources, try MDAP. For non-standard or irregular speaker arrangements, use DBAP.

The algorithm dropdown only appears for surround formats. For Ambisonics output, the algorithm is always Ambisonics. For binaural output, HRTF convolution handles spatialization directly.

4.6 HRTF Profiles

When Binaural output is selected, the **HRTF Profile** dropdown offers six options. Each profile uses different measured head-related transfer functions, producing a subtly different spatial sound:

Profile	Source	Character	CPU
Simple (Low CPU)	Woodworth head model	No convolution — ITD+ILD only, lowest latency	Minimal
Immersive	SADIE II D2 KU100	Rich spatial detail, strong elevation cues	Normal
Natural	CIPIC Subject 003	Organic, realistic binaural rendering	Normal
Precise	HUTUBS PP2	Detailed, analytical spatial accuracy	Normal
Spatial	Bernschuetz KU100	Dense full-sphere measurement, widest coverage	Normal
Studio Reference	MIT KEMAR Large Pinna	Neutral, classic reference standard	Normal

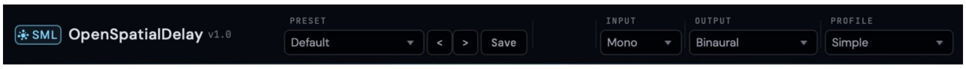
Choosing a profile

> Start with Studio Reference for mixing. Switch to Simple if you need lower CPU usage.
>
> Different profiles suit different head shapes. Try each to find the one that sounds most natural to you — what works best is personal.

Low-frequency bypass: Some HRTF measurements lack low-frequency content below 200 Hz. OpenSpatialDelay automatically detects this and routes bass directly to the output, bypassing the HRTF convolution. This ensures solid bass regardless of which HRTF profile you choose.

5 Controls Reference

5.1 Header Bar



The header bar contains: plugin title and version (left), **preset navigation** (dropdown, prev/next arrows, save button), **OSC toggle** (ADM-OSC receive on/off), **Output Format** dropdown, and either **Algorithm** (surround) or **HRTF Profile** (binaural) dropdown, which swap automatically based on the selected format.

5.2 DELAY Section

Controls the core delay timing and feedback. Accent color: **cyan**.

Parameter	Range	Default	Description
TIME	1–2000 ms	500 ms	Base delay time. Tap k reads at k × TIME.
SYNC	On / Off	Off	Locks delay time to DAW tempo
NOTE	1/32 – 2/1	1/4	Note division (when SYNC is on)
MODE	Notes, Dotted, Triplet	Notes	Sync modifier (when SYNC is on)
FEEDBACK	0–100%	30%	Amount of delay output fed back to input

5.3 MOD Section

Delay-time LFO modulation (wobble). Accent color: **rose**.

Parameter	Range	Default	Description
MOD toggle	On / Off	Off	Enable wobble modulation
AMOUNT	0–100%	0%	LFO modulation depth (milliseconds)
MORPH	0–100%	0%	Waveform shape: sine → triangle → square

5.4 TONE Section

Feedback path filters and air absorption. Accent color: **violet**. Filters apply to the feedback path only — the first echo is always unfiltered.

Parameter	Range	Default	Description
FLT toggle	On / Off	Off	Enable feedback path filters
HP	20–5000 Hz	50 Hz	High-pass cutoff frequency
LP	200–20000 Hz	5000 Hz	Low-pass cutoff frequency
HP RES	0.1–8.0	0.71	High-pass filter resonance (Q)

Parameter	Range	Default	Description
LP RES	0.1–8.0	0.71	Low-pass filter resonance (Q)
AIR toggle	On / Off	Off	Distance-based HF rolloff per tap

5.5 MIX Section

Output mixing and gain staging. Accent color: **amber**.

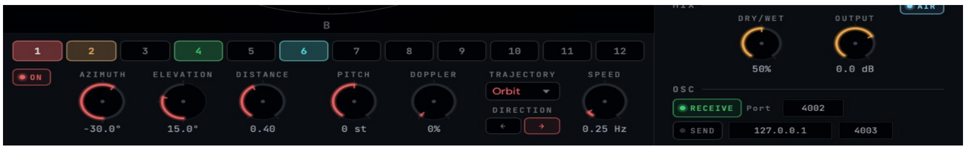
Parameter	Range	Default	Description
DRY/WET	0–100%	50%	Blend between dry input and processed output
IN	-96 to +24 dB	0 dB	Input gain (wet path only, not dry signal)
OUT	-96 to +24 dB	0 dB	Output gain (applied to both dry and wet)

5.6 OSC Section

ADM-OSC receive and send controls. Accent color: **green**. See ADM-OSC for protocol details.

Parameter	Default	Description
RECV toggle	Off	Enable ADM-OSC receive (incoming position control)
Port	4002	UDP port for incoming OSC messages
SEND toggle	Off	Enable ADM-OSC send (broadcast positions)
IP	127.0.0.1	Target IP for OSC send
Send Port	4003	Target port for OSC send

5.7 Per-Tap Controls (Bottom Panel)



Select a tap by clicking its numbered button (1–12). The controls below update to show that tap's settings:

Parameter	Range	Default	Description
ON/OFF	Toggle	Taps 1–4: On	Enable/disable this tap
AZIMUTH	-180 to +180°	Varies	Horizontal angle (clockwise knob = clockwise map)
ELEV	-90 to +90°	0°	Vertical angle
DIST	0.1–3.0	0.5	Distance from listener (affects attenuation)

Parameter	Range	Default	Description
DOPPLER	0–100%	0%	Doppler effect intensity for moving taps
PITCH	-12 to +12 st	0 st	Per-tap pitch offset (adds to global pitch)
INPUT	L+R / L / R	L+R	Which stereo input channel feeds this tap
TRAJ	14 shapes	None	Trajectory animation shape
SPEED	0.00–5.00 Hz	0.30 Hz	Trajectory animation speed
DIR	Fwd / Rev	Fwd	Trajectory direction (forward or reverse)

6 Trajectories & Animation

Each tap can follow an animated trajectory through 3D space. Select a shape from the TRAJ dropdown in the bottom panel.

Origin-point architecture: The knob values (azimuth, elevation, distance) set the tap's origin point. The trajectory animates around this origin. You can reposition a running trajectory by adjusting the knobs — the animation follows. A crosshair marker on the spatial map shows the origin position.

Shape	Motion	Best For
None	Static position, no animation	Manual placement, static scenes
Bounce	Azimuth and elevation oscillate back and forth	Vertical bouncing motion
Circle	Horizontal orbit at constant elevation and distance	Smooth circular panning
Cross	Alternates between azimuth and elevation axes	X-pattern motion
Figure-8	Two-tangent-circles pattern in azimuth/elevation	Elegant weaving motion
Heart	Cardioid-shaped path around the origin point	Expressive, organic arcs
Helix	Spiral with rising/falling elevation over time	3D corkscrew motion
Infinity	Horizontal figure-8 (lemniscate) path	Smooth side-to-side weave
Line	Back-and-forth along a single azimuth axis	Pendulum-like motion
Orbit	Circular path around the listener at constant height	Rotating delays, circular ping-pong
Random	Multi-sine noise on all axes (never repeats)	Organic, unpredictable movement
Spiral	Outward spiral with distance variation	Expanding/contracting patterns
Square	Rectangular path rotated by base azimuth	Angular, geometric motion
Triangle	Triangular path rotated by base azimuth	Sharp directional changes

Speed controls the animation rate (0.00–5.00 Hz). **Direction** can be Forward or Reverse. Each tap animates independently — mix different shapes and speeds for complex spatial motion.

If [ADM-OSC receive](#) is active and an external controller sends position data for a tap, the trajectory pauses for that tap. It automatically resumes 500ms after the last OSC message.

Creative tip

Try setting 4 taps on Orbit at different speeds (0.5×, 1.0×, 1.5×, 2.0×) for a mesmerizing rotating delay pattern. With feedback above 50%, the trails create evolving spatial textures.

7 Creative Tips

OpenSpatialDelay can go far beyond basic delay effects. Here are some techniques to explore:

Self-Oscillation

Set Feedback above 95% to create self-oscillating delay trails that sustain indefinitely. The built-in output limiter (soft saturation at +2 dB) prevents digital clipping.

For analog-style oscillation, enable the TONE filter and set LP to 2–3 kHz. The delay will sing with a warm, filtered tone. Adjust FEEDBACK to control the balance between sustain and decay.

Pitch Spiraling (Shimmer)

Set global PITCH to +12 semitones (+1 octave). Each tap accumulates pitch: tap 1 = +1 octave, tap 2 = +2 octaves, and so on. Each feedback cycle adds another round-trip of pitch.

For subtler shimmer, try +5 or +7 semitones (a fourth or fifth). Combined with high feedback and LP filtering, this creates lush, cathedral-like textures.

Wobble — Tape Effects

Enable MOD and set AMOUNT to 15–25% with MORPH at 0% (sine wave). This creates a subtle tape-like wow and flutter on the delay.

Higher MORPH values (toward 100%) produce more rhythmic, square-wave modulation — useful for choppy, glitchy delay effects.

Stereo Width with Input Routing

On stereo tracks, try setting alternating taps to L and R input channels: Tap 1 = L, Tap 2 = R, Tap 3 = L, and so on.

This creates a natural stereo ping-pong enhanced by spatial positioning. Position L-channel taps on the left side of the spatial map and R-channel taps on the right for maximum width.

Feedback Filter Design

The TONE section filters apply to the feedback path — each time the signal recirculates, it passes through the filters again. Over successive echoes, this creates progressive tonal shaping.

High-pass at 200–400 Hz removes bass buildup in long feedback tails. Low-pass at 3–5 kHz

> simulates tape machine degradation. Combine both for a focused, telephonic character that opens up beautifully in 3D space.

8 Presets

OpenSpatialDelay includes **70 factory presets** across 9 categories: Classic Delays, Spatial Movement, Ambient + Texture, Height + 3D, Surround Production, Wobble + Modulated, Creative + Experimental, Rhythmic, and User.

Browse presets using the dropdown in the header bar. Use the **‹ › arrows** to step through presets, or click the preset name to open the full categorized menu.

8.1 Preset File Locations

macOS: `~/Library/Audio/Presets/OpenSpatialDelay/`

Windows: `%APPDATA%\OpenSpatialDelay\Presets\`

Factory presets are written to this directory on first launch. User presets are saved as JSON files in the same location.

8.2 Saving Presets

Click the **Save** button in the header to open the save overlay. Enter a name and press Return to save, or Escape to cancel.

9 ADM-OSC Integration

OpenSpatialDelay supports **ADM-OSC** (Audio Definition Model over Open Sound Control), the industry-standard protocol for controlling spatial audio object positions in real time.

9.1 Receive (Incoming Control)

Enable **RECV** in the OSC section and set the port (default 4002). External tools like SPAT Revolution, L-ISA Controller, or custom scripts can send position data to control tap positions in real time. Supported messages include `/adm/obj/N/azim`, `/adm/obj/N/elev`, `/adm/obj/N/dist`, `/adm/obj/N/aed` (combined), and `/adm/obj/N/xyz` (Cartesian, auto-converted to polar).

9.2 Send (Broadcast Positions)

Enable **SEND** and set the target IP and port (default 127.0.0.1:4003). The plugin broadcasts tap positions as `/adm/obj/N/aed` messages at 30 Hz, with position-change gating to minimize network traffic. Use this to sync with external spatial renderers.

10 Troubleshooting

Issue	Cause	Solution
No sound output	All taps disabled	Enable at least one tap (click ON button in bottom panel)
Only dry sound, no echoes	Dry/Wet at 0% or format mismatch	Set Dry/Wet above 0%. Ensure track channel count matches selected Output Format.
Binaural doesn't sound 3D	Wrong HRTF profile for your hearing	Try different HRTF profiles. Start with Studio Reference.
Audio crackling/dropouts	Buffer size too small or too many HRTF taps	Increase DAW buffer to 256+ samples. Switch to Simple profile for lower CPU.
Plugin not appearing in DAW	Installation path issue	Verify plugins are in ~/Library/Audio/Plug-Ins/ (macOS). Rescan in DAW.
Delay not syncing to tempo	Tempo Sync is off	Enable the SYNC toggle in the DELAY section
Runaway self-oscillation	Very high feedback + resonant filters	The output limiter prevents clipping, but reduce Feedback or filter resonance if too intense.
OSC not connecting	Port conflict or wrong number	Check that no other app uses the same port. Verify port matches sender/receiver.

OpenSpatialDelay v1.0

Part of the Spatial Media Library Suite

Designed by Andrew Rahman · Architecture by Claude

spatialmedialab.org

github.com/AndrewRahman/OpenSpatialDelay

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